

CE 524P : Computational Hydroclimatology Lab

Course Code : CE 524P

Course Name : Computational Hydroclimatology Lab

L-T-P-C : 0-0-2-1

Intended for : PG and Ph.D. students

Prerequisite : None

Mutual Exclusion:

Approval: 56th BoA

List of Experiments

1. **Acquiring the hydroclimatic data:** Acquisition and preprocessing of hydroclimatic data from various sources (2 Hours)
2. **Analysis of Climate Change Trends:** Analyze long-term temperature and precipitation data to identify and quantify climate change trends. (2 hours)
3. **Handling Missing Hydroclimatic Data:** Apply various techniques to handle missing data in hydroclimatic time series and evaluate their effectiveness. (2 hours)
4. **Simple Energy Balance Climate Model:** Implement a zero-dimensional energy balance climate model to simulate global temperature changes under different forcing scenarios. (2 hours)
5. **Hydrologic Modelling with Lumped Models:** Set up and calibrate multiple lumped models to characterize the hydrologic behavior of the watershed.(2 Hours)
6. **Hydrologic Modeling with SWAT:** Set up and run a SWAT model for a small watershed to simulate hydrological processes and assess impacts of land use changes. (2 hours)
7. **Downscaling and Bias Correction of Climate Data:** Apply statistical downscaling and bias correction techniques to GCM outputs for use in local impact studies. (2 hours)
8. **Analysis of Climate Indices:** Calculate and analyze various climate indices to understand large-scale climate patterns and their local impacts. (2 hours)
9. **Drought Analysis using Standardized Precipitation Index (SPI):** Compute SPI at multiple timescales to analyze historical drought patterns and assess future drought risk. (2 hours)
10. **Flood Frequency Analysis:** Perform flood frequency analysis using annual maximum streamflow data and assess non-stationarity due to climate change. (2 hours)
11. **Climate Change Impact on Crop Yield:** Assess the impact of climate change on crop yield using a simple crop model and climate projections. (2 hours)
12. **Glacier Mass Balance Modeling:** Implement a simple glacier mass balance model to assess climate change impacts on glacier evolution. (2 hours)

Text Books:

1. None. Experiment notebooks will be provided during the lab



Reference Books:

1. None. Experiment reference material will be provided during the lab